



**Politecnico
di Torino**

DET
GUIDING ELECTROMAGNETIC SYSTEMS

Project 18 Report

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1 Design

The objective of this project was to design a maximally flat (Butterworth) low-pass microstrip filter: based on the specifications, the filter must have a cutoff frequency of $f_c = 2.4 \text{ GHz}$ and provide a minimum stopband attenuation of 25 dB at $f_s = 4.8 \text{ GHz}$.

The filter is designed on a substrate with a relative permittivity of $\varepsilon_r = 4.45$, a loss tangent of $\tan(\delta) = 0.0188$, and a thickness of $h = 1.55 \text{ mm}$.

1.1 Prototype and Filter Order

To meet the out-of-band attenuation requirement at $\omega_s/\omega_c = 2$, the necessary filter order N for a Butterworth response was calculated: an order of $N = 5$ ensures an attenuation greater than 30 dB at 4.8 GHz , satisfying the 25 dB constraint, and the corresponding normalized low-pass prototype values $(g_1, g_2, \dots, g_{N+1})$ were extracted from standard maximally flat filter tables.

1.2 Stepped-Impedance Implementation

The low-pass prototype was implemented using the stepped-impedance technique, in which the normalized prototype elements (series inductors and shunt capacitors) are approximated by alternating high and low impedance microstrip transmission line sections.

As dictated by the project guidelines, the highest available impedance ($Z_{high} = 120 \Omega$) was utilized for the inductive sections, the lowest one ($Z_{low} = 15 \Omega$) was utilized for the capacitive sections, and a standard impedance ($Z_0 = 50 \Omega$) was utilized for the input and output access lines.

Using microstrip synthesis equations for the specified substrate, the physical trace widths were calculated.

The electrical lengths ($\beta\ell$) for each section were computed to yield the required inductive or capacitive reactances, and subsequently converted into physical lengths.

2 Simulation

The synthesized physical dimensions were initially modeled and analyzed using AWR Microwave Office, like shown in Figure 1.

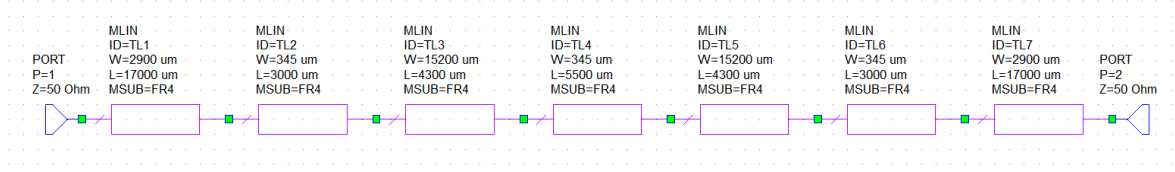


Figure 1: AWR schematic representation of the stepped-impedance low-pass filter.

Due to the parasitic reactances introduced by the microstrip step discontinuities, the initial frequency response exhibited a slight shift in the cutoff frequency.

An optimization routine was performed on the segment lengths to precisely restore the -3 dB cutoff to 2.4 GHz while preserving the deep stopband rejection at 4.8 GHz.

Following schematic tuning, an electromagnetic simulation (EM) was conducted using AXIEM (Figure 2) to verify the complete physical layout.

This simulation accurately captures fringing fields, coupling between adjacent segments, and radiation losses.

The EM-simulated scattering parameters successfully verified the design criteria.

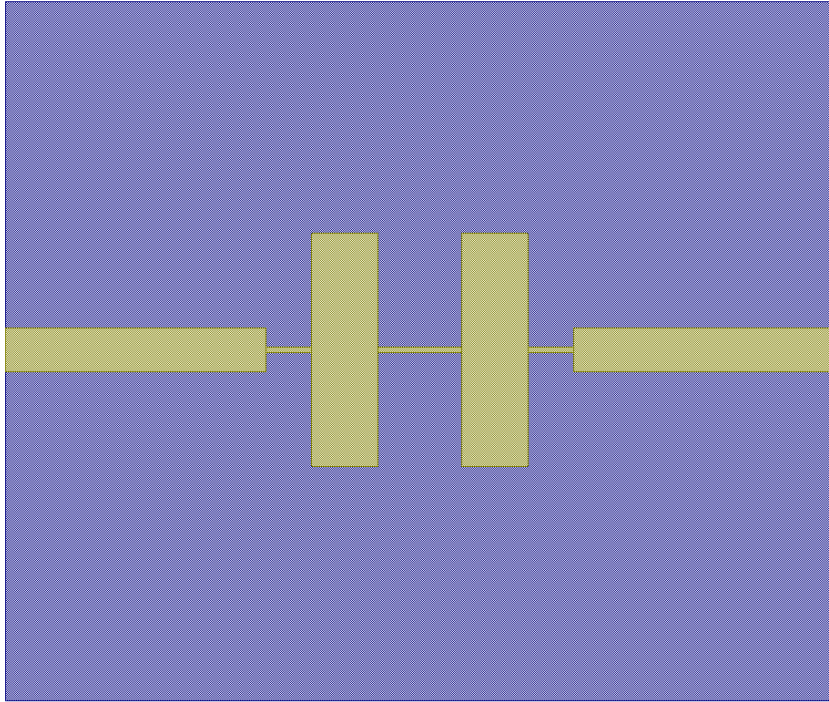


Figure 2: AXIEM EM simulation layout of the tuned microstrip filter.

3 Characterization

The optimized filter layout was exported to a GERBER file and manufactured.

To assess the physical performance of the prototype, the manufactured circuit was characterized using a Keysight P9375A Vector Network Analyzer, whose interface is shown in Figure 3.

Standard 50 Ω SMA connectors were mounted to the input and output access lines to interface with the analyzer.

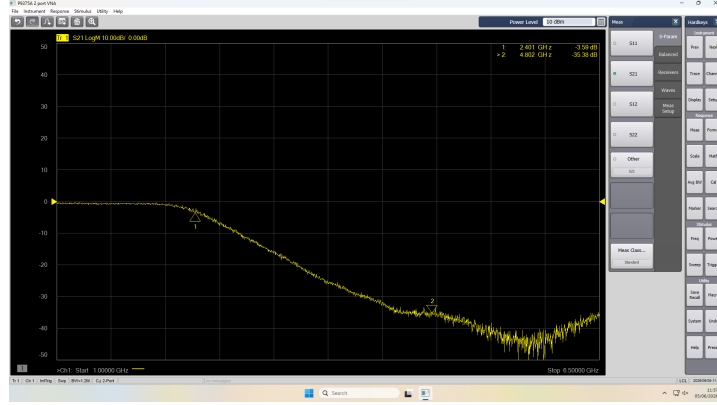


Figure 3: Measurement setup using the Vector Network Analyzer.

The physical S -parameter measurements (recorded in Touchstone $.s2p$ format) were imported into MATLAB and overlaid directly with the AWR simulation data to evaluate the accuracy of the design, as shown in Figures 4 and 5.

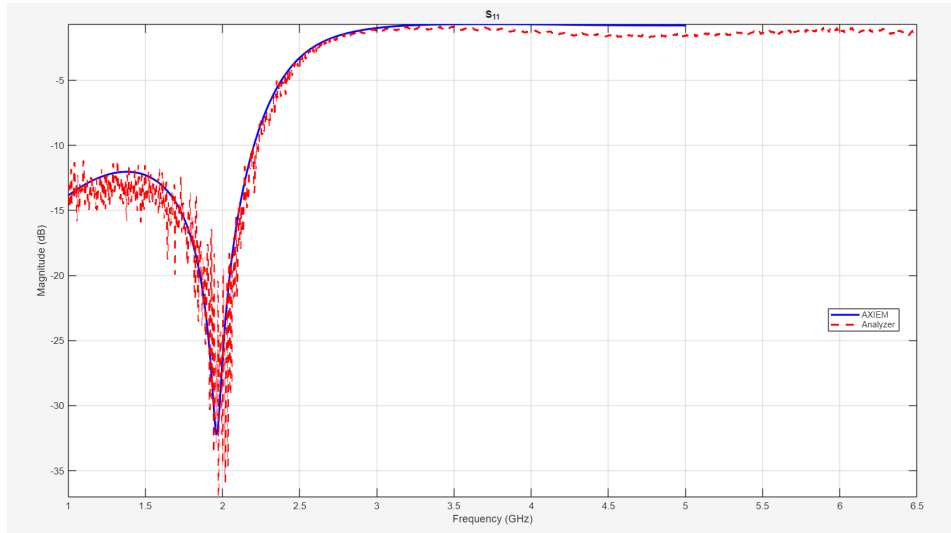


Figure 4: Comparison of Simulated vs. Measured S_{11} (Reflection Coefficient).

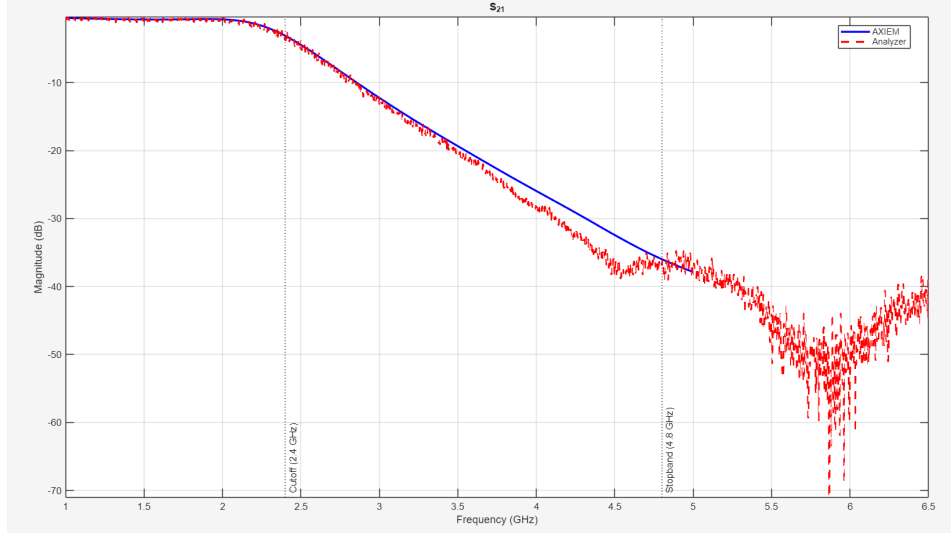


Figure 5: Comparison of Simulated vs. Measured S_{21} (Transmission Coefficient).

3.1 Discussion of Results

The physical measurements show strong agreement with the AWR EM simulations.

Figure 5 compares the theoretical and measured S_{21} responses, showing a maximally flat low-pass behavior: the response crosses the 3 dB threshold at the targeted 2.4 GHz cutoff frequency, while the attenuation reaches approximately 35 dB at 4.8 GHz, safely exceeding the 25 dB specification.

Minor discrepancies are observable between the simulated and measured curves, particularly a slight increase in insertion loss in the passband and a marginal shift in the ripple phases. These deviations can be attributed to several practical factors:

- **Manufacturing Tolerances:** Slight over-etching or under-etching of the copper traces during fabrication alters the physical widths and lengths, impacting the characteristic impedances of the extreme 15 Ω and 120 Ω lines.
- **Connector Parasitics:** The soldering of the SMA connectors introduces unmodeled parasitic inductances and capacitances at the input and output ports.
- **Substrate Variations:** The relative permittivity ($\epsilon_r = 4.45$) and loss tangent of standard circuit board materials are nominal values and can fluctuate slightly over wide frequency bands.

Overall, the stepped-impedance microstrip filter was successfully synthesized, optimized, and verified to meet all project specifications.

4 Appendix

4.1 Impedance Calculation

```
clear; clc; close all;

fc = 2.4e9;
wc = 2 * pi * fc;
Z0 = 50;

Z_high = 120;
Z_low = 15;

g = [0.618, 1.618, 2.000, 1.618, 0.618];
N = length(g);

fprintf('\n>>> CONFIGURATION 1: Pi-Network (Capacitor-First)
<<<\n\n');

for k = 1:N
    if mod(k, 2) ~= 0
        C = g(k) / (Z0 * wc);
        beta_l_deg = rad2deg(g(k) * (Z_low / Z0));

        fprintf('Section %d: Shunt Capacitor\n', k);
        fprintf('  Target Z      : %d Ohms\n', Z_low);
        fprintf('  Lumped Value : %.3f pF\n', C * 1e12);
        fprintf('  Elect. Length: %.2f degrees\n\n',
            beta_l_deg);
    else
        L = (g(k) * Z0) / wc;
        beta_l_deg = rad2deg(g(k) * (Z0 / Z_high));

        fprintf('Section %d: Series Inductor\n', k);
        fprintf('  Target Z      : %d Ohms\n', Z_high);
        fprintf('  Lumped Value : %.3f nH\n', L * 1e9);
        fprintf('  Elect. Length: %.2f degrees\n\n',
            beta_l_deg);
    end
end

fprintf('\n>>> CONFIGURATION 2: T-Network (Inductor-First)
<<<\n\n');
```

```

for k = 1:N
    if mod(k, 2) ~= 0
        L = (g(k) * Z0) / wc;
        beta_l_deg = rad2deg(g(k) * (Z0 / Z_high));

        fprintf('Section %d: Series Inductor\n', k);
        fprintf('  Target Z      : %d Ohms\n', Z_high);
        fprintf('  Lumped Value : %.3f nH\n', L * 1e9);
        fprintf('  Elect. Length: %.2f degrees\n\n',
            beta_l_deg);
    else
        C = g(k) / (Z0 * wc);
        beta_l_deg = rad2deg(g(k) * (Z_low / Z0));

        fprintf('Section %d: Shunt Capacitor\n', k);
        fprintf('  Target Z      : %d Ohms\n', Z_low);
        fprintf('  Lumped Value : %.3f pF\n', C * 1e12);
        fprintf('  Elect. Length: %.2f degrees\n\n',
            beta_l_deg);
    end
end

```

4.2 Analysis

% Compare AXIEM measurements with Analyzer

```
clear; clc; close all;
```

%% Import AXIEM Data

% Read S11 AXIEM data

```
opts11 = detectImportOptions('S11_measurements_AWR.txt');
```

```
opts11.DataLines = [2 Inf];
```

```
awr_s11_data = readmatrix('S11_measurements_AWR.txt', opts11
    );
```

```
f_awr_s11 = awr_s11_data(:, 1);    % Frequency in GHz
```

```
s11_awr_dB = awr_s11_data(:, 2);   % Magnitude in dB
```

% Read S21 AXIEM data

```
opts21 = detectImportOptions('S21_measurements_AWR.txt');
```

```
opts21.DataLines = [2 Inf];
```

```
awr_s21_data = readmatrix('S21_measurements_AWR.txt', opts21
    );
```

```
f_awr_s21 = awr_s21_data(:, 1);    % Frequency in GHz
```

```

s21_awr_dB = awr_s21_data(:, 2);    % Magnitude in dB

%% Import Analyzer Measurement Data
meas_obj = sparameters('S_Analyzer.s2p');

% Convert to GHz
f_meas = meas_obj.Frequencies / 1e9;

% Extract S11 (1,1) and S21 (2,1) and convert to dB
s11_meas_lin = squeeze(meas_obj.Parameters(1, 1, :));
s11_meas_dB = 20 * log10(abs(s11_meas_lin));

s21_meas_lin = squeeze(meas_obj.Parameters(2, 1, :));
s21_meas_dB = 20 * log10(abs(s21_meas_lin));

%% Plot S11 (Reflection Coefficient)
figure('Name', 'S11 Comparison', 'Position', [100, 100, 800,
    500]);
plot(f_awr_s11, s11_awr_dB, 'b-', 'LineWidth', 2);
hold on;
plot(f_meas, s11_meas_dB, 'r--', 'LineWidth', 2);
grid on;
title('S_{11}');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
legend('AXIEM', 'Analyzer', 'Location', 'best');
axis tight;

%% Plot S21 (Transmission Coefficient)
figure('Name', 'S21 Comparison', 'Position', [150, 150, 800,
    500]);
plot(f_awr_s21, s21_awr_dB, 'b-', 'LineWidth', 2);
hold on;
plot(f_meas, s21_meas_dB, 'r--', 'LineWidth', 2);

% Add vertical markers for the specific Project 18 design
requirements
xline(2.4, 'k:', 'Cutoff (2.4 GHz)', 'LabelVerticalAlignment',
    'bottom', 'HandleVisibility', 'off');
xline(4.8, 'k:', 'Stopband (4.8 GHz)', 'LabelVerticalAlignment', 'bottom', 'HandleVisibility',
    'off');

grid on;

```

```
title('S_{21}');  
xlabel('Frequency (GHz)');  
ylabel('Magnitude (dB)');  
legend('AXIEM', 'Analyzer', 'Location', 'best');  
axis tight;
```